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# Automating MARC Holdings Generation with AI: Lessons from the Field

**Built with an AI coding assistant; runs without one.**

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More detail, the figures and a PDF of this poster
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## 1 The problem, and the approaches I tried

### The problem

Many libraries' serials holdings are stored only as human-readable, free-text **866** fields rather than as MARC-encoded **853/863** pairs. Converting by hand is slow, conventions vary, and hand-typed data is prone to typos and other errors.

> **The goal:** a tool that is portable across collections, works on messy holdings as they are (no manual clean-up), and never silently loses anything.

866 · as a person typed it

```
v. 27 no. 4-v. 31 no. 4
a  a  b  b  a  a  b  b

(April 1992-April 1996)
 j     i     j     i
```

| subfield | 853 caption | 863 held |
|---|---|---|
| **$a** volume | v. | 27-31 |
| **$b** issue | no. | 4-4 |
| **$i** year | (year) | 1992-1996 |
| **$j** month | (month) | 04-04 |

```
853 31 $8 1 $a v. $b no. $i (year) $j (month)
863 40 $8 1.1 $a 27-31 $b 4-4 $i 1992-1996 $j 04-04
```

APPROACH 1

### Write the script by hand

I adapted a script by USC colleagues (Chung & Pendyala, 2023) over six months while teaching myself Python. The hardest part was a single, very long regular expression.

APPROACH 2

### Ask a model to read the statements

Send the 866s to a large language model through its API, and ask it to pick out the volumes, issues and dates.

APPROACH 3 · CHOSEN

### Use AI to build the tool

Describe in plain language what the tool should do, and have an AI coding assistant (Claude Code) write a deterministic Python application.

| | 1 · BY HAND | 2 · A MODEL READS | 3 · AI BUILDS THE TOOL |
|---|---|---|---|
| **Coding knowledge** | The most | The least | Some |
| **Getting started** | Over six months | Minimal prompting | A first working tool from one prompt |
| **Cost** | Nothing | Tokens for every statement | About $20/month (Pro plan) to build; nothing to run |
| **Same output every time** | Yes | Not guaranteed | Yes |
| **Other collections** | Poorly. It needs clean, consistent 866s | Not tested far enough | Designed for it |
| **Understanding the code** | Fully, line by line | No code to read | Partly: a black box, checked by tests |

## 2 The approach chosen: AI-assisted development of a deterministic tool

### How the Workbench reads a statement

1 · Cut into labeled pieces

```
v.  27  no.  4   -   v.  31  no.  4
VOL  #  ISS  #   —  VOL  #  ISS  #

(  April  1992  -  April  1996  )
     CHRON  YEAR  -  CHRON  YEAR
```

2 · Its shape

```
VOL ISS — VOL ISS(CHRON YEAR-CHRON YEAR)
```

```
v. <start_vol> no. <start_iss>-v. <end_vol> no. <end_iss>
(<start_month> <start_year>-<end_month> <end_year>)
```

1. Statements with the same shape are one group, whatever their numbers.
2. Each group gets a matching rule with a named slot for each value.
3. A cataloger confirms what each value means, once per group.
4. The parser reads every statement; a pattern fills only what it cannot know.

### What the cataloger controls

- **On a pattern:** what each value is, its caption and level; skip a shape; test or edit the rule; reuse the library on another collection.
- **On one record:** edit an 866 before it is read, split it, set this record's 853 (frequency, numbering, issues per volume, missing captions), skip it, leave a note.
- **For the whole file:** holdings level, subfield convention, strict reading, what to do with a value MARC has no place for, and whether to keep each 866.

Every decision is listed in a downloadable log.

### The working rules

1. **Measure before you change.** Count first, then fix.
2. **Nothing disappears silently.** Encoded, named, or held.
3. **One reader of holdings.** Patterns only fill its gaps.
4. **If a fix needs the tests rewritten, stop.** Rethink the fix.
5. **Write the changelog for catalogers,** not for coders.

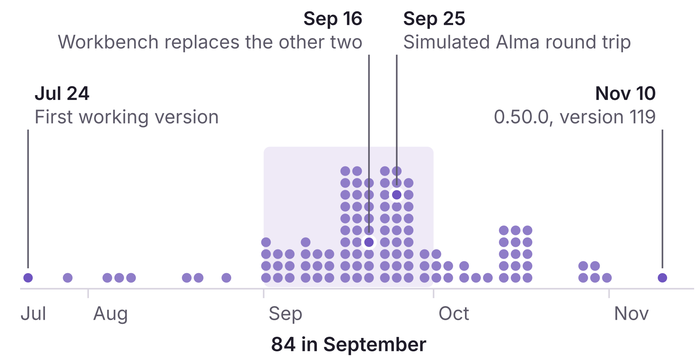
### The human in the loop

The AI writes the code; I decide what "correct" means. Using the tool, I found two bugs the tests had missed:

- `Late Summer 2002` was coded as plain Summer, but there may be an Early Summer issue too. Now flagged (0.9.6).
- `v. 58 Suppl. (Sep 2003)` lost "Suppl." when its pattern was confirmed. Supplements are now held (0.32.0).

### How it was built

119 numbered versions in 15 weeks, most of them small.




Each dot is one version. The Pattern Detector and Converter each began as a single prompt. The Workbench combined them, then replaced them on September 16.

### On real data

A real Alma export of 372 records:

**1,057** holdings statements

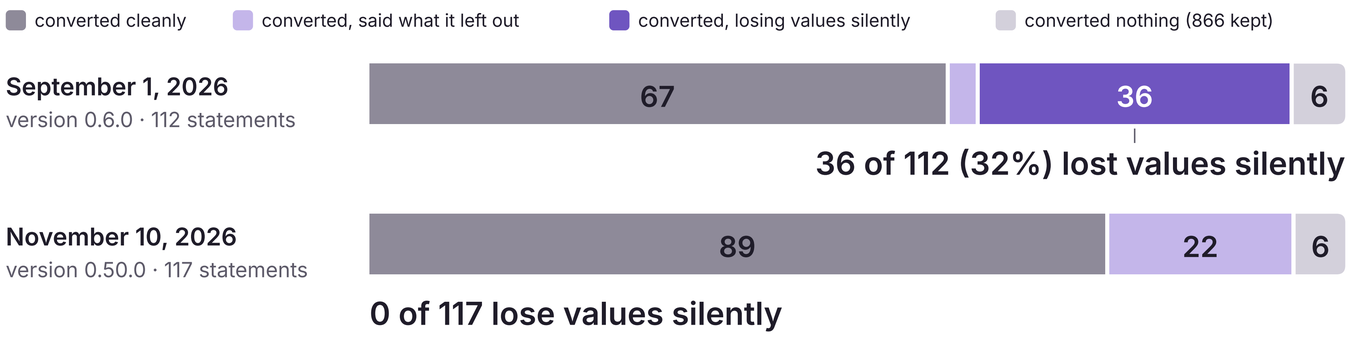
**0** values unaccounted for

**0** drift, simulated Alma round trip

**48/48** of Alma's own 866s reproduced

### The turning point: silent losses

A silent loss writes the fields, removes the 866, and drops a volume or a month with nothing on screen to say so.




Zero is what was measured, not a guarantee. A review by a new AI model, Fable 5.2, found one more: `v.3 (1990--1995)` kept only 1990. Fixed in 0.50.0.

### Lessons, and what's next

- **A script reads what it was written for and passes over the rest.** So the first rule is that nothing disappears silently: every value is encoded, named as left out, or held with its 866 kept.
- **Decide whether you can work with code you don't fully understand.** Tests show the tool works, not that it was built well. That takes a software engineer to judge.
- **Where this may be heading:** people build their own tools with AI, and engineers build the guardrails.
- **The models improved as I worked.** Newer ones solved what had stopped older ones, then built features from one prompt.
- **AI is at its best building the tool, not being it.**
- **Next:** more real .mrc files from colleagues, a review by a software engineer, then a 1.0 release.

**Try it:** tools.matthewcodey.com/workbench · **Source:** github.com/bdmcodey/holdings-workbench
**Contact:** codey@usc.edu (work) · matthew.codey@gmail.com (personal)

Chung, M., & Pendyala, P. C. (2023). Enhancing serials holdings data: A Pymarc-powered clean-up project. *Code4Lib Journal*, (58).